

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: KWAN Tsz Hei Marcus

Student ID: 22012026D

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

Command: "show ip route ospf"

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

The interface s0/0/0 in R2 is configured to advertise OSPF routes as it sets "no passive-interface".

The interface s0/0/1 in R2 remained as a passive interface, so no OSPF routes are advertised there.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

Command: "config t" -> "router ospf 1" -> "no passive-interface s0/1/1" -> "end"

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65, it is calculated based on the distance between R3 and the network 192.168.2.0.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To better match the speed of the Ethernet of each link, so that the path of OSPF routers can be better and accurately determined.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost of 192.168.3.0 is calculated from adding 1 (from R3's G0/0) to 781 (from R1's S0/0/1).

The cost of 192.168.23.0 is calculated from adding 781 (from R1's S0/0/1 or S0/0/0) to 64 (from R2 or R3's S0/0/1).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562, because as the bandwidth of the link R2-R3 is set to 128, the cost of such a link increased from 64 to 781. So, the result becomes 781 (from R1's S0/0/1 or S0/0/0) + 781 (from R2/R3's S0/0/1)

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the cost of R1's S0/0/1 increased to 1565, so the cost for routing to 192.168.3.0 via R3 is 1565.

However, the cost for routing to the same subnet via R2 (i.e., R1's S0/0/0) is only 1564, calculated from 781 (from R1's S0/0/0) + 781 (from R2's S0/0/1) + 1 (from R3's G0/0), which is lower than via R3.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Because if no router ID is assigned to a router, the network will derive the router ID from the highest active IP address on the router's interface. Moreover, the router ID may be changed by the system if any router interface is disconnected.

2. Why is the DR/BDR election process not a concern in this lab?

Because all hosts in the network are connected to a single router only, i.e., a stub network. Also, unlike a transient network, all routers are connected point-to-point in the whole area (i.e., there is no "central" point), so no router serves as the DR.

3. Why would you want to set an OSPF interface to passive?

Since setting an OSPF interface to "passive" reduces stress on that interface, it frees up bandwidth for other network traffic. This is helpful if the network grows larger, as routing updates may contain a large amount of information to be sent over router interfaces, which can consume significant bandwidth.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The continuous updating of the OSPF routing table makes it hierarchical. Since OSPF is a link-state vector-based protocol, every router has its own converged routing database. Even when a new router connects to the existing network, it will reach convergence using "Hello" and routing database update packets, and the new router will build its own routing database.

areas

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Command: "network 10.0.0.0 0.255.255.255 area 0"

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP, the one with the highest metric, will be used for the routing decision.